

MAMACO'S WEB BASED BOARDING HOUSE AND DORMITORY MANAGEMENT SYSTEM

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INTRODUCTION

Boarding houses are establishments that lodgers pay for one or more rooms to be rented out each night, sometimes for weeks, months, or even years. Some services, such as laundry and cleaning, may be provided, and the common areas of the house are maintained. The guest can rent for one or more nights and sometimes for a longer period. Anent this, Mamaco's Boarding-house is one of the popular boarding houses in Kabacan. The building is situated in Plang Village 3, Kabacan, North Cotabato. The area is strategically located near the University of Southern Mindanao Kabacan, North Cotabato. Also, different renters are residing in Mamaco's Boarding-house. Some of them are employees, and most boarders are students from schools within the Poblacion. Indeed, most of the boarders are students from the University of Southern Mindanao.

Hence, the researcher will conduct this study to create technologies for assessing the current processes used in managing boarding houses. As observed, most boarding houses use the manual approach in managing their establishments. It is documented that manual operation will slow business operations, which affects the quality of lodging among the guests. Moreover, the researcher accumulated information and discovered that lodging proprietors experience difficulties in overseeing their boarding houses. Furthermore, based on the researcher's observation of Mamaco's boarding house management, the management has concerns about boarders' issues, clients' inquiries, and payments status so that the

researcher will develop technology for boarding-house and dormitory management for easy transaction and proper management of the establishment.

Statement of the Problem

Currently, the boarding house owner does not have an automated or computerized system in managing his boarding house business. Thus, the management uses the traditional means to operate the transactions and processes in his boarding house. Hence, there are concerns about the efficiency of using traditional methods in operating the boarding house. According to the manager, it is difficult for him to record and generate information on boarders, rooms status, and payments. The manager cannot produce reports in time, which causes a delay in transactions. Whenever there is an inquiry regarding the availability of rooms, payment, booking, and room reservation, the manager cannot answer automatically since data and information are mismanaged. To solve the issues, the researcher would like to propose the development of Mamaco's web-based boarding house and dormitory management system.

Specifically, the system will allow monitoring boarders' payment, the status of rooms and beds (available or occupied), expenses bills, and generation of reports. Consequently, the system will improve boarding houses' operations by eliminating the manual processes and errors encountered in the pre-existing

management system. Indeed, the system will make boarding house and dormitory management easy, fast, convenient, accurate, and reliable.

Objectives of the Project

The general objective of this system is to design and develop a Ministry of Social Services and Development Informatics for Dr. Richard M. Albutra General Hospital. Specifically, it aims to:

1. develop a system that will collect necessary data and information of the clients;
2. develop a system to monitor the availability of the funds in MSSD-BARMM;
3. develop a system that will monitor the clients who availed assistance once within a three-month period;
4. develop a system that will provide downloadable clients documents, such as General Intake Sheet, Certificate of Eligibility, and Endorsement Letter, and
5. evaluate the system in terms of functionality.

Significance of the Project

The result of this study will be beneficial to the following:

Boarding House/Dormitory Owners - The system will help improve their business efficiency and allow them to cope up with technological trends in managing boarding houses.

Landladies/Landlords - The system will allow them to easily and efficiently track and monitor boarders' payment, the status of rooms and beds (available or occupied), payments, and generation of reports. They can conveniently and efficiently manage the whole operation of the boarding house and dormitory.

Boarding House/Dormitory Tenants/Boarders - The system will improve their overall experience in staying in the boarding house. They can personally access the records of their payments as well as electronically relay their concerns and complaints about the boarding house.

Researcher - Their experience in conducting the project will further enhance their knowledge and skills as researchers.

Future Researchers - The study can serve as their guide if they wish to design and develop their version of the Boarding House and Dormitory Management System.

Expected Outputs of the Project

The project entitled *Momaco's Web-Based Boarding House and Dormitory Management System* will be designed and developed using web technologies such as PHP, My SQL, and Bootstrap. Since the project is browser or web-based, it means that it will work online, and it can also be implemented using offline settings. Moreover, the system will provide a solution to the needs of the boarding house and dormitory. Also, this endeavor will develop a technology to improve the operation of the boarding house and dormitory by eliminating the manual processes with its errors.

Operational Definition of Terms

The following are terms used in the study:

Web-Based – refers to a software program that is accessed over a network connection using HTTP rather than existing within a device's memory that is used through a web browser.

Management System – refers to the system that provides a well-organized and guaranteed record-keeping system with minimum space and effort needed.

Entity Relationship Diagram – refers to an abstract and conceptual representation of data.

Use Case Diagram –refers to a list of steps, typically defining interaction between roles and systems to achieve the goal.

Acceptability - refers to the point in the lifecycle at which every aspect of the application is developed.

Effectiveness - refers to the measure of the extent to which a system is expected to achieve a set of specific mission requirements expressed as a function of availability, dependability, and capability.

Functionality– refers to the software's capability to provide functions and implement users' needs.

Productivity - refers to a set of guidelines or processes designed to help users get things done.

Customers – refers to the clients of the boarding house.

Conceptual Framework

This figure shows the flow of the project. Specifically, the input is the system's users, which refer to the admin/owner and customers. All users must input their email and password to log in to the website. Then, the client can inquire about service availability and prices. Next, the admin can input payments of all registered boarders. Admin can process client inquiries and monitor payments, room availability, and data collection. The admins can edit data on payments and view and add boarders to the list. Thus, the admin can view the boarders' profile and reports. Also, both the admin and boarders can view their records and receipts, which serve as projects' output. It will improve the operational management of the boarding house and dorm. In a gist, the project's outcome will result in the efficient management of the boarding house. Specifically, the admin or system's users will efficiently render services on billing, payments, and record keeping.

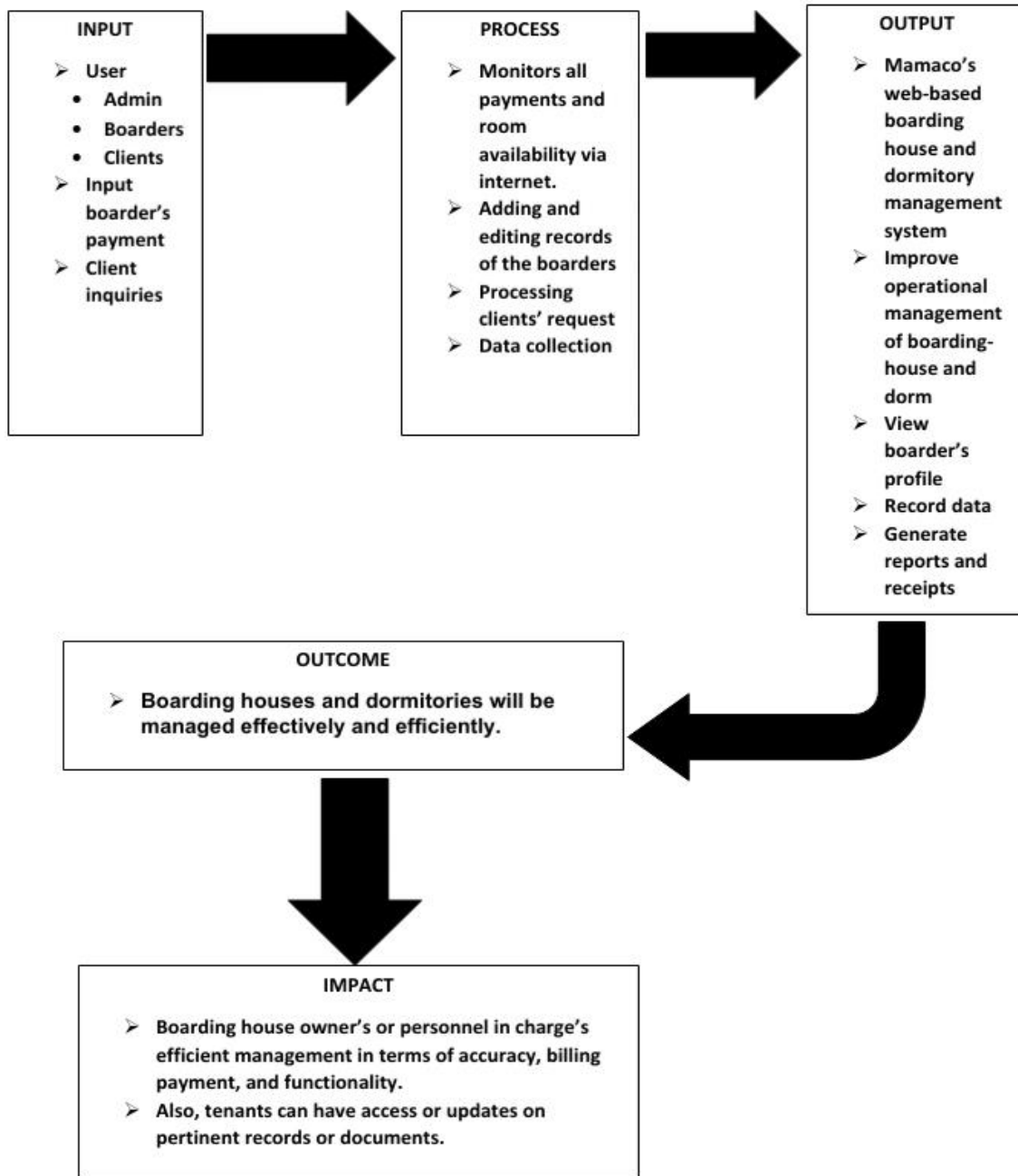


Fig. 1. Conceptual framework of Mamaco's Web-Based Boarding House and Dormitory Management System.

Limitations of the Project

This study will focus on the development process of a web-based boarding house and dormitory management. Specifically, the researcher will design and develop a boarding house and dormitory management system that will seamlessly streamline processes of managing boarding houses. To accomplish her goals, the researcher will gather boarders' data and personal documents, such as ID, name, contact, rooms, address, and payments. Then, she will obtain the total summary of reports of all the remittance incurred every month and for the whole year. Consequently, the system will categorize the data payments on a yearly and monthly basis. On the other hand, the owner or employee in charge can access the system. Moreover, the system can generate a receipt to validate payment transactions. In addition, the framework illustrates that tenants can transact through GCash. The client can also process an online inquiry on room availability.

This study will limit the data on boarders' personal information, payment records, rental costs, and room availability. Only the owner or admin and the employee in charge can access the system. Furthermore, the notification of cleaners, miscellaneous reports, the announcement of activities and events are not included in the system's function.

REVIEW OF RELATED LITERATURE

Boarding houses are offered in many academic institutions. They offer non-resident transitory settlements, especially for students. The establishment is anticipated to be conducive to learning and a secure dwelling place. However, boarding houses in the Philippines, in general, need to upgrade their services (LL Ortiz, 2012).

Foreign-Related System

Online Dorms Systems allows customers to book their dorm room from anywhere in the world. This is an automated system that allows users to check the availability of dorm rooms. The dates can be used to narrow down the search. The available rooms have the status of being available. It will show you all the rooms that are available as of the search date. The user can cancel the reservation within 48 hours after it has been made, and there is a concept of user login. The user can log into his account after creating it with his email identity card. The administrator who controls the dormitories system program has access to an administrator login page. Boarding house owners can also look at the profit and expansion of the company. The proposed technique saves time and effort for both students and dorm administrators. The goal of this application is to provide an online dorm system for improving software quality and reliability that can be used for internal applications.

This approach can be used to reduce student problems with an application/module, assign dorms to individuals, and solve student problems (Mothe, KK, Soraguni, R. & Vakity, 2010).

In boarding schools, an Innovative Web-based Dormitory Information System was developed. The purpose of establishing the SMA Terpadu Krida Nusantara is to integrate the three pillars, namely: Academic, Religion, and Vocational skills. The technology adjusts to the development of information technology. SMA Terpadu Krida Nusantara wants to utilize information technology as a medium to assist the process of boarding management so that it will become a web-based board management information system, which later the information system can be accessed by students' parents from their homes. However, the system has limitations. Specifically, the web-based dormitory management information system for SMA Terpadu Krida Nusantara boarding school includes making student input data in/out of the weekly excursion, input data on student loyalty and neatness in the dormitory, input data for student counseling guidance in the dormitory, and input data for sick students. SMA Terpadu Krida Nusantara wants to use information technology to help with boarding management to become a web-based board management information system that parents can access from their homes. Moreover, the PHP programming language and MySQL database were used to construct the web-based dormitory administration information system for SMA Terpadu Krida Nusantara boarding school. Making student input data in/out of the weekly excursion, input data on student loyalty and neatness in the dormitory,

input data for student counseling guidance in the dormitory, and input data for sick students are all part of the Web-based dormitory management information system for SMA Terpadu Krida Nusantara boarding school. The RUP (Rational Unified Process) technique was utilized in this system, which was based on the UML (Unified Modeling Language) model. The findings of this study are expected to result in an application that aids in the management of dorms at SMA Terpadu Krida Nusantara. Therefore, there are research objectives, such as using the PHP programming language and the MySQL database and constructing a web-based dormitory administration information system for the SMA Terpadu Krida Nusantara boarding school. Also, the system is a breakthrough in the construction of a web-based dormitory management information system for the boarding school SMA Terpadu Krida Nusantara. It also enables to use of RUP (Rational Unified Process) to create an information system for the SMAT-KN boarding school's management. The findings of this study may aid in making the process of data management more efficient and straightforward, allowing for speedier reporting of student data (Jafrudin Jafrudin & Yeffry Handoko Putra, 2020).

The boarding school monitoring system is intended to make it easier for both parents and wardens to keep track of their children's activities. Both the hardware and software components are required to create this system. The system is set up to help students, parents, and wardens in the best way possible. To begin, the student or parent had to fill out an application for an outing activity or an overnight activity on the internet. After the application has been successfully filed,

both the student and the parent must wait for a response from the system, which will indicate whether the application has been approved or rejected. The teacher or warden has complete control over the response. If the application is approved, the system can move on to the next level (Sarnin, Ya'acob, Azize, Yusof, & Sarnin, 2018).

Local Related System

The dormitory management system is an information system that can replace manual processing and monitoring of students in the dormitory. The mentioned project was developed in Visual Basic and MySQL. The following bibliography is intended as a guide and reference for executing and developing a customized version of investigator bedroom management that meets the needs and requirements of the client (Tsavo Media Canada Inc., 2019).

The Boarding-house Management System is a system that monitors tenant payments, the status of rooms and beds (available or occupied), water and electricity bills, and much more. Conventionally, these administrative areas were processed manually. Most pensions use the manual approach to manage the above areas. The main functions of the pension management system are focused on the payment records of the tenants, the monitoring of the status of the rooms and beds, the encryption of the tenant information, the generation of reports, and the copying of database security. The system is developed according to the software

development life cycle (SDLC) technique. SDLC technology focuses on the interplay of planning, analysis, and design tasks leading to implementation, followed by operation and support. The survey and observation of current processes and methods were used as primary data collection techniques. Pension owners, landlords, and tenants participate as participants in the study. With the introduction of the system, pension management becomes effective and efficient. Manual processes and errors are eliminated, which improves the functioning of pensions. The implementation of the system makes pension management easy, fast, convenient, accurate, and reliable (Django, 2020).

Good data management, such as a Web-based Boarding School Monitoring System, can and will save someone's life in the event of an unwelcome situation. A simple, accurate, and reliable data record for boarding school pupils can and will simplify the parent-teacher monitoring system. The traditional approach, on the other hand, requires students to personally notify and record their log to their parents and teachers. There is a lack of automation in the system, which could lead to various issues. Inaccurate logging time, a misplaced outing card, and an unstructured log record are all issues. To overcome this problem, the Web-Based Boarding School Monitoring System (WEBMOS) was created, which uses a MySQL database and RFID technology. This technology provided a means for accurately recording student log data. The goal is to create a system that allows parents and teachers to access correct data with a single click on their devices. This system employs RFID module technology to track the activity of students' records in real-

time. Arduino Mega, Ethernet Shield, Global System for Mobile Communication (GSM), and Radio Frequency Identification (RFID) Module are used in the monitoring system. After the student flashes their card to the RFID reader, the system runs an automated data log, the data is posted to the database, and the uploaded information may be seen on the webpage at any time and from anywhere. As a result, the system can collect and logging precise real-time data logging for student activity (Ya'acob, Azize, Yusof, Sarnin, Naim, & Rohaizad, 2018).

The booking system is a systematic record of information and transactions. Contains the lists of all customer records for future use. Provides permanent evidence and sources of records of recorded transactions. An online reservation system is a computerized system that stores and distributes hotel information and helps hoteliers improve online sales and marketing. In addition, it increases information processing, repeats critical processes, and provides accurate results. This system also secures all reservation transactions that would lead to customer satisfaction. Booking and reservation systems refer to the registration of the names and other information of its clients. Computerizing your reservation system helps the manager and employees do their jobs effectively (Django, 2018). According to Delizo (2013), the online room reservation system is efficient and brilliant software, which is nevertheless easy to use. The system grants full authority and power when reserving hotel or motel rooms over the Internet. As a result, all guest-in payments can be accumulated. They can enter their room descriptions, facilities, prices, and assignments in the reservation system. It also allows users to

confirm ownership in real-time on the hotel website and easily complete the sale. The system computerizes the reservation process, which helps to carry out many transactions at a low cost. It allows one to check available inventory and complete a reservation online making the reservation process more efficient and less time-consuming. OHRS supports room guests and agents with various payment options, such as credit/debit cards. The system can regularly track hotel performance as all payment information is updated online and emailed to the reservation manager.

METHODOLOGY

Research Design

Descriptive research design will be used to determine the user's acceptance, effectiveness, functionality, and productivity of Mamaco's Web-Based Boarding House and Dormitory Management System.

Research Participants and Materials

The researcher respondents will be the prospected admin or user or manager and the clients or boarders of Mamaco's Boarding-House and Dorms. The researcher will use a self-made questionnaire to gather necessary data and information.

Data Collection

Initially, a request letter to conduct the study will be sent to the boarding house's owner. Once approved, a series of interviews will be conducted to acquire relevant data and information, which will be used in formulating the study. Observation of the establishment's operations are completed will be done. The

researcher will distribute questionnaires to identified respondents to assess whether the system is functional or not. The respondents will be the twenty-eight randomly selected borders, the owner, or the manager. Moreover, additional data and information relevant to the study will be gathered from related readings, either online or printed materials.

Data Analysis

This part of the chapter presents the data collected from the respondents of the cited boardinghouse and dormitory and its owner. Specifically, the owner and clients of the boarding house will be study respondents. The researcher will need at least fifty respondents, which will comprise 49 tenants, and (1) the owner of Mamaco's Boarding House and Dormitory to determine the efficiency of the proposed system.

Table 1: Data Analysis

Numerical value	Description	Weighted range
1	Yes	10-5
2	No	5-1

Validity

The project is a system that allows boarding house and dorm management to electronically monitor and track the operation of the boarding house. Anent this, the system will serve as an online platform for recording and transacting between clients and boarding house' management. PHP and Bootstrap will be used to develop the said template. Mamaco's Web-based Boarding House and Dormitory Management System is a database-driven system that will automate monitoring boarder's payment, the status of rooms and beds, water and electricity bills, and generation of reports.

Moreover, automated management is a system that will enable the electronic management of boarding houses. Also, the Records Management is a database system that makes the records of the boarding house's operation electronic, safe, accurate, reliable, and fast. Furthermore, Report Generation can generate reports about the payments, bills, and status of rooms and beds in a boarding house.

Ethical Consideration

The researcher will adhere to ethical considerations. The researcher will search the names of her participants through social media. Then, she will chat with them individually to fully inform them regarding the purpose of the study. She will assure that their responses will be treated with the utmost confidentiality. The researcher will also assure the respondents that the data to be used in this study is for academic purposes only.

Furthermore, the researcher will ask permission before she commences her interview. Indeed, the researcher will ensure that she will create and maintain a conducive environment among her respondents.

Project Development Approach

The Iterative and Incremental Waterfall Model shows that the system will be utilizing the SDLC model for Mamaco's Web-Based Boarding House and Dormitory Management System. Thus, it will be helpful to the concerned individuals because they can focus on every step from problem identification to the acceptance stage. Indeed, this concept is an easier and more understandable cycle process. Also, this kind of model is suitable for the project. Her research participant voluntarily participated in the study. Also, she will assure her respondents that all the data to be collected will be kept confidential.

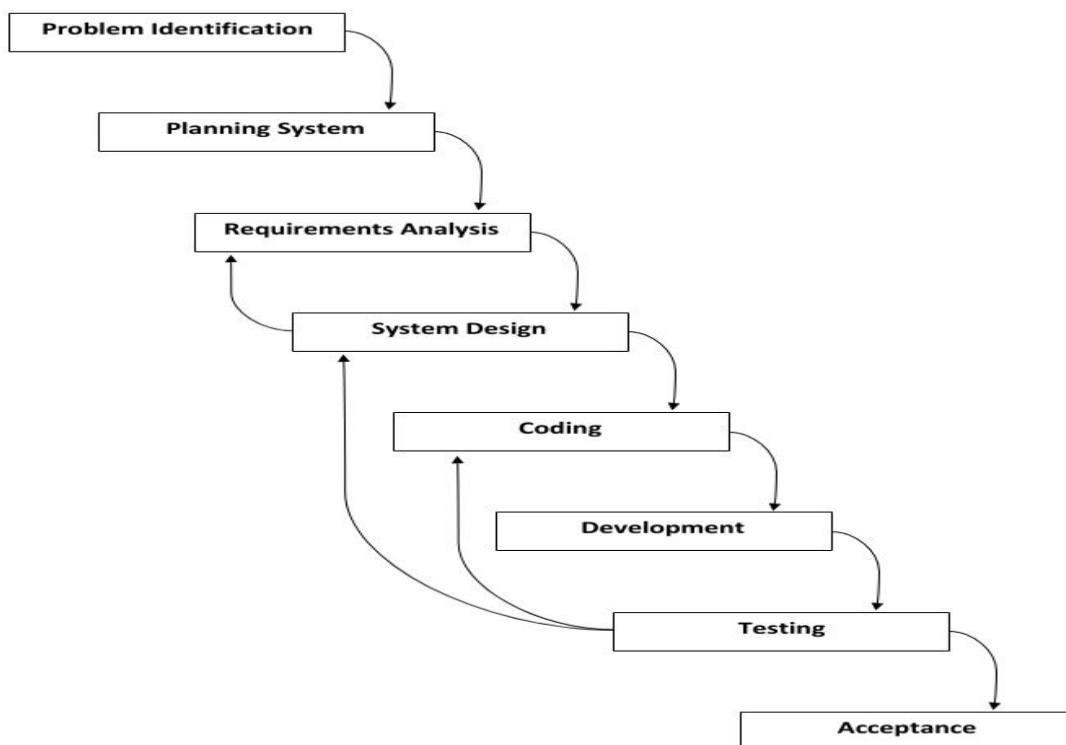


Fig. 2. Waterfall Model of Mamaco's Web-Based Boarding House and Dormitory Management System.

Problem Identification

This is the first and initial phase of the waterfall model. The phase is used to identify the problem.

Planning System

At this stage, the researcher schedules the time for modeling, coding, testing, and software development. The researcher also estimates the cost of each phase in this stage.

Requirements Analysis

In this phase, the researcher gathers requirements and thinks about the solution to a problem.

System Design

System design is a blueprint of the solution. The process follows the steps first to create logical design and physical design. If during the design phase it is noticed that there are some more requirements for designing the code, the analysis

phase is revisited. Then, the design phase is carried out according to the new set of resources.

Coding

In this stage, the logical models are constructed by developing small programs, which are also called units. These small program units are integrated to form the software.

Development

In the Development phase, the researcher starts coding and developing the software. This is the longest phase of the waterfall model as developers need more time to build the software. Once the development of the software is completed, then the project is handed over to the testers.

Testing

Testing is a process to validate the system's objective through various steps. It is a mechanism that makes sure that the system is working properly or not. If there are any flaws, the software development process must step back to the design phase. In the design phase, changes are implemented and then the succeeding stages of coding and testing are again carried out. If the researcher still does not bite the desired development, it will possibly turn into coding to code again, and then to make sure that the client should not face any problem while installing and using the software.

Acceptance

This is the last stage of the software development in the waterfall model. Proper execution of all the preceding stages ensures an application based on the provided requirements. Most importantly, it ensures a satisfied client.

Project Development

The following figure shows the Gantt chart illustrating the tasks to be accomplished by the researcher. It also reflects the time allotted for the establishment of the entire system. Figure 2 shows the activities or steps taken for the proposed project completion. It shows the task name, duration, starting date, and end date.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	G	A	N	T	T			C	H	A	R	T		
2	Task Name					Duration		Start Date			End Date			
3	1. Planning Project Phase					43 Days		March 13, 2021/Saturday			May 03, 2021/Monday			
4	1.1 Generate the Problem					7 Days		March 19, 2021/Friday			March 26, 2021/Friday			
5	1.2 Produce the PROJECT Schedul					10 Days		March 27, 2021/Saturday			April 06, 2021/Tuesday			
6	1.3 Conduct Interviews					13 Days		April 07, 2021/ Wednesday			April 20, 2021/Tuesday			
7	1.4. Lunch the Project					13 Days		April 21, 2021/Wednesday			May 03, 2021/Monday			
8	2. Analysis Phase					43 Days		May 04, 2021/Tuesday			Jun 10, 2021/Saturday			
9	2.1 Problem Analysis					10 Days		May 26, 2021/Wednesday			June 05, 2021/Saturday			
10	2.2 Gather Data					20 Days		Jun 06, 2021/Sunday			June 26, 2021/Saturday			
11	2.3 Define System Requirements					5 Days		Jun 07, 2021/Sunday			July 01, 2021/Thursday			
12	2.4 Prioritise System Requirements					8 Days		July 02, 2021/Friday			July 10, 2021/Saturday			
13	3. Define Phase					167 Days		July 11, 2021/Sunday			January 02, 2022/Sunday			
14	3.1 Working on Chapter 3					95 Days		July 12, 2021/Monday			October 15, 2021/Friday			
15	3.2 System Methodology Modeling					6 Days		October 16, 2021/Saturday			October 22, 2021/Friday			
16	3.3 Data Flow Modeling					6 Days		October 23, 2021/Saturday			October 28, 2021/Thursday			
17	3.4 Diagram Modeling					6 Days		October 29, 2021/Friday			November 04, 2021/Thursday			
18	3.5 Design the User Interface					7 Days		November 05, 2021/Friday			November 12, 2021/Friday			
19	3.6 Design and Integrate Database					23 Days		November 13, 2021/Saturday			December 06, 2021/Monday			
20	3.7 Prototype for Design Details					7 Days		December 07, 2021/Tuesday			December 14, 2021/Tuesday			
21	3.8 Design and Integrate System Conyrol					10 Days		December 15, 2021/Wednesday			December 25, 2021/Saturday			
22	3.9 Design the System Interface					7 Days		December 26, 2021/Sunday			January 02, 2022/Sunday			
23	4. Implementaion Phase					116 Days		January 03, 2022/Sunday			May 19, 2022/Thursday			
24	4.1 Contrast Software Components					39 Days		January 04, 2022/Monday			February 12, 2022/Saturday			
25	4.2 Test and Verify					30 Days		February 13, 2022/Monday			March 15, 2022/Tuesday			
26	4.3 Convert Data					17 Days		March 16, 2022/Wednesday			April 17, 2022/Sunday			
27	4.4 Document and Train					18 Days		April 13, 2022/Monday			May 08, 2022/Friday			
28	4.5 Install the System					12 Days		May 09, 2022/Saturday			May 19, 2022/Thursday			

Fig.3. The Gantt Chart of Mamaco's Web-Based Boarding House and Dormitory Management System.

Database Schema

Figure 4. shows the database schema of the user namely: User id, username, password, contact. The customer table has customer id, username, fname, mname, lname, address, age, gender, civil status, and contract. Also, the room table has room id, room number, room name, description, bed number, room type. Then, the room type table has room id, type name. Moreover, the bed table has bed id, bed number, room id, and price. Furthermore, the booking table has a booking id, bed id, customer id, start date, end date, total amount, user id. Lastly, the payment table has payment id, payment amount, date from, date to, user id, paid by, remark, payment status, and the guardian table has guardian id, guardian name, contact, address, and customer id.

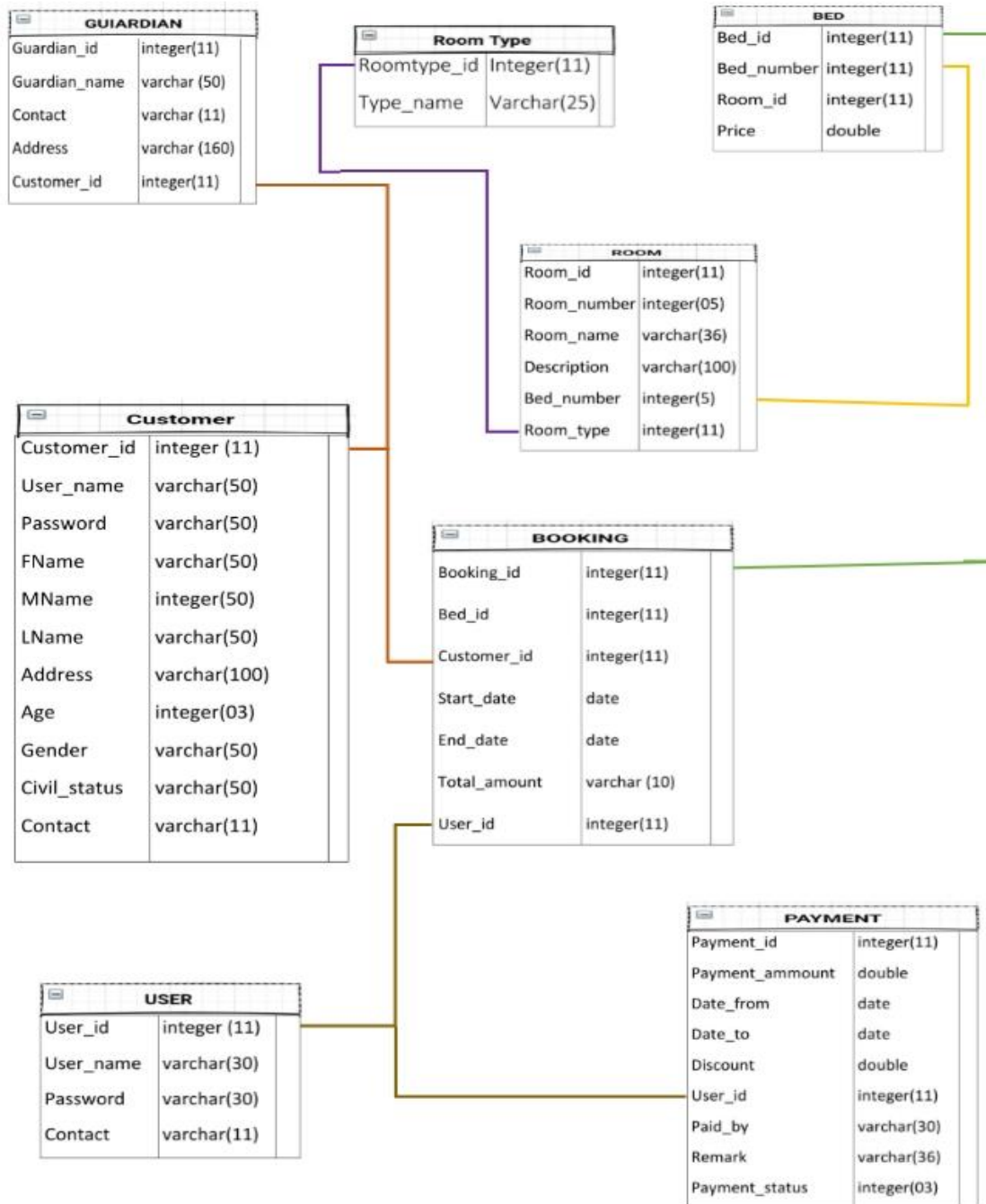


Fig. 4. Shows the database schema of Mamaco's Web-Based Boarding House And Dormitory Management System.

System Requirement Specification

This part was the document or set of documentation that describes the features and behavior of a system. It is a structured collection of information that embodies the requirements of Mamaco's Web Based Boarding House and Dormitory Management System.

Functional Requirements

1. Log In

Description: The system provides a facility to log in to the system.

Input – Enter Username and Password

Processing – The system will check the input of the user. If the entries are valid, then the user can log in. Otherwise, the user will be asked to re-enter the username and password.

Output – User Profile Page

2. Inquiry

Description: It is an act of a client asking for information and details about the establishment.

Input– Clients' requests for services.

Processing – Mamaco's Web-based Boarding House and Dormitory Management System Process request.

Output: View the list of availability and non-availability services.

3. Registration

Description: To be part of the boarding house and dormitory automated system, the customer should register him/herself first. The customer should put all information about him/her that includes the chosen room and bed.

Input –Enter personal information

Processing –The system will add the customer to the boarders' list with its data and information.

Output – Boarder first record.

4. Add Boarder to List

Description: The employee in charge adds the customer as a boarder or member of the boarding house. The boarder's record like personal information, room/bed, and payments will be viewed by the employee in charge.

Input – Put client name into boarder's list.

Processing –Saving the boarder's data and starting day.

Output –Boarder records.

5. View Boarder's Details

Description: The employee in charge can view all boarders' data and information.

Input –Visit document.

Processing –Viewing boarders list with each data and information.

Output –Monitored record.

6. Edit Payments

Description: The employee in charge can access to edit the boarder's payment. To get the final amount of the boarder payments, the employee in charge should review the boarder utility cost, start, and end date, discounted, or added payments. This part will be shown the payments status of the boarder if she/he has still a balance/credit or is settled.

Input –Input price of boarder paying utility cost from start to due date. Then, enter boarder's balance or paid payments and put discount or additional payments.

Processing –Prompts payments amount.

Output –View payments status.

7. Manage Boarder's Record

Description: employee in charge is going to manage boarder's record like when to start and due date, status, and payment amount. All details that the in charge need to manage about boarders.

Input – View boarder's record.

Processing – Showing boarders personal information guardian, information, transaction date, and payment status.

Output – Record.

Description: In Report Module (RM), users will find all reports that clients have access to transaction data, including payments status, end or start date, and even the user's utilization of the system.

Input –Generate report.

Processing –Transacting data based on date.

Output –View report module output.

8. View Records

Description: This will be used to view all transacted data.

Input –Select boarder's name.

Processing – Showing boarder's personal information, guardian information, transactions date, and payment status.

Output –Pooled documents.

9. Print Receipt

Description: Printed receipt is a hard copy of the record that shows the amount of all payment's utilities, amount of discount or deduction, amount of additional or added payments, balance amount, the total amount of payments, and transaction date of payments. Here, the users can view who paid and failed to pay.

Input –Setup

Processing –Printing document.

Output –Receipt output.

10. Log Out

Description: Logging out means ending access to a website. It signifies the system that the current user wishes to end the login session.

Input –Select logout.

Processing –The system will process the logging out of the user.

Output –Display logout.

Software Requirements Specification

This table shows the software requirements that will be used by the developer for the application design

Table 1. Software Requirements Specification

SOFTWARE	MINIMUM	RECOMMENDED
Operating System	Windows 2000/XP	Intel Core i5 (sixth generation or newer) or equivalent
Internet Browser	Internet Explorer 8	Internet Explorer 9
Apache 4.0	Apache 4.0	512 MB RAM recommended

Hardware Requirements Specification

This table shows the computer hardware that the developer will use to develop the project

Table 2. Hardware Requirements Specification

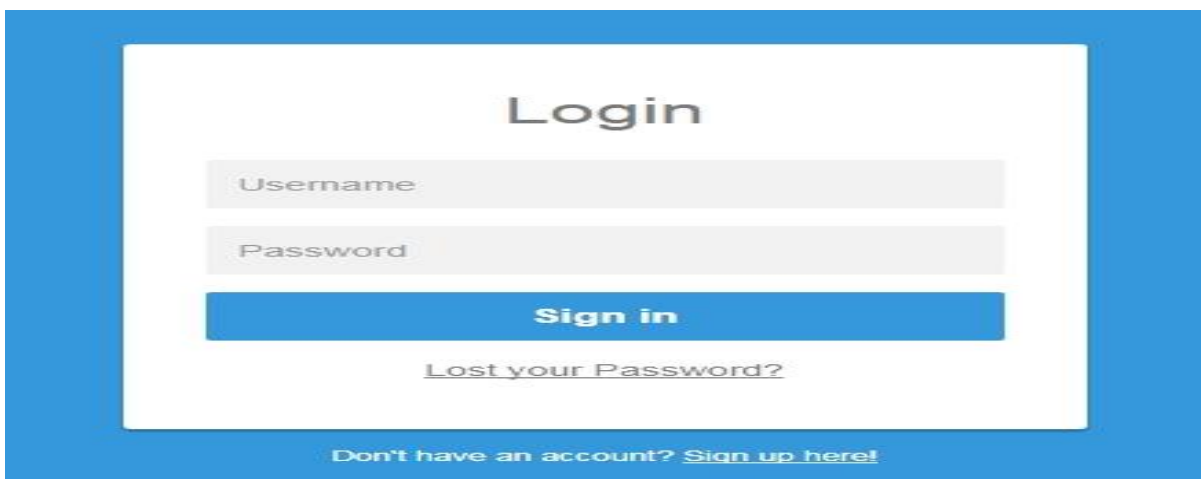
SOFTWARE	MINIMUM	RECOMMENDED
Processor	1,6 GHz	2 x 1,6 GHz CPU
RAM	1,75 GB RAM	3,5 GB RAM
My SQL	4GB	6GB
Back End	4GB RAM	8GB RAM

Project Design

The project design will be developed using web technologies, such as PHP, My SQL, and Bootstrap. The web application will be developed under Internet development, and Microsoft SQL Server Management Studio will be used as a database development application.

User Interface Design

Figure 5. Shows the login when operating Mamaco's Web-Based Boarding House and Dormitory Management System. In this form, the system's users must input the right username and password to access the system. This will require the users to input the correct combination of username and password to login into the system. The users will set unique usernames and passwords that they will use to gain access to the records and features of the system.



The image shows a login form with a blue background. The form is white and contains the following elements:

- Login** (Title)
-
-
- Sign in** (Button)
- [Lost your Password?](#)
- [Don't have an account? Sign up here!](#)

Fig. 5. The Login Form.

Figure 6. The dashboard is the homepage of the system administrator's

Administrator. The admin can access and manage records found in the dashboard.

The dashboard mainly displays the following information: Number of Boarders, Number of Rooms, and Number of Beds

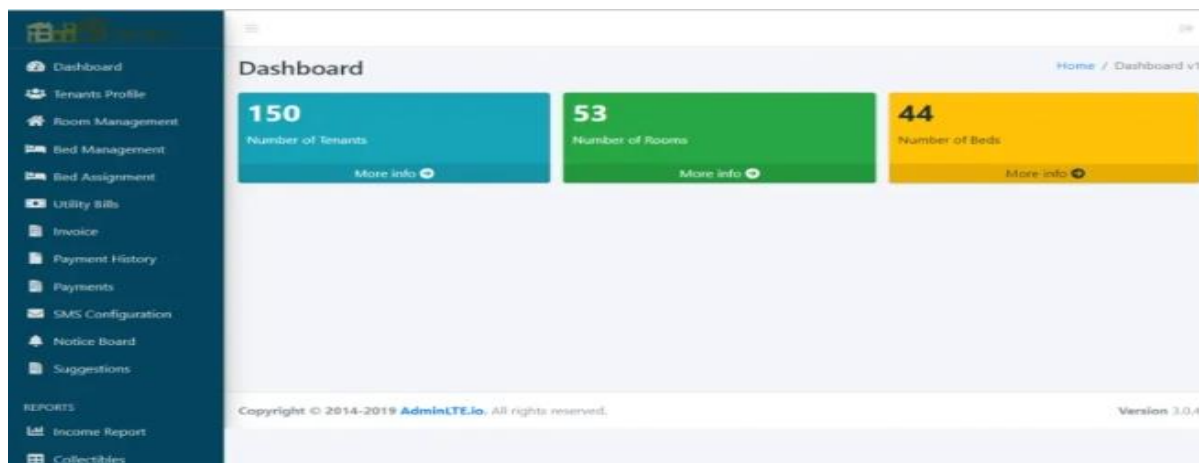


Fig. 6. Dashboard.

Figure 7. allows encoding information of the tenants. Thus, all boarding house tenants' records will be entered into the system using the form. The following information will be encoded in the system: Last name, first name, Middle name,

Complete Address, Email Address, Contact Information, Gender, Profile Picture, and Proof of Identity – upload.

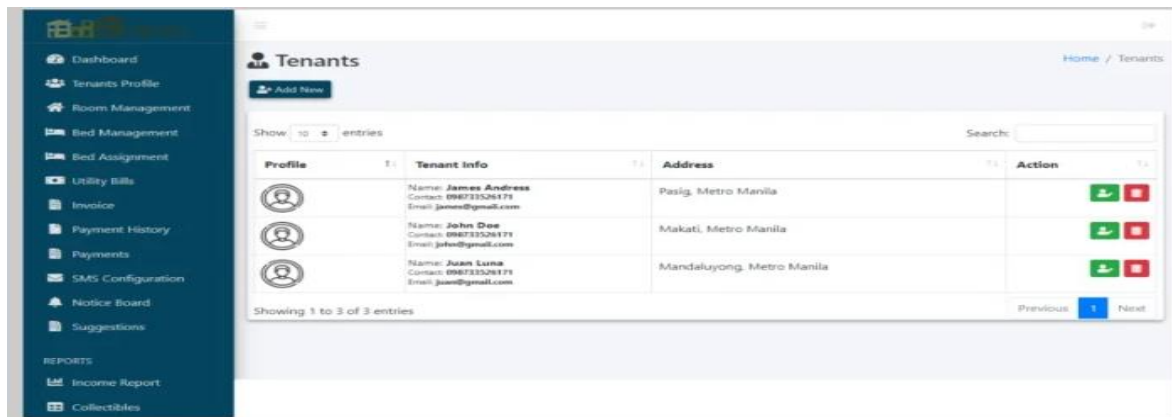


Fig. 7. Boarder's profile form design.

Figure 8. displays the income collection of the boarding house and dormitory.

Month	Income
September	Php 5000.00
October	Php 5000.00
November	Php 5000.00
December	Php 5000.00
Total Income	Php 20,000.00

Fig. 8. Income Reports.

Figure 9. Shows the design of the Submit Proof of Payment form.

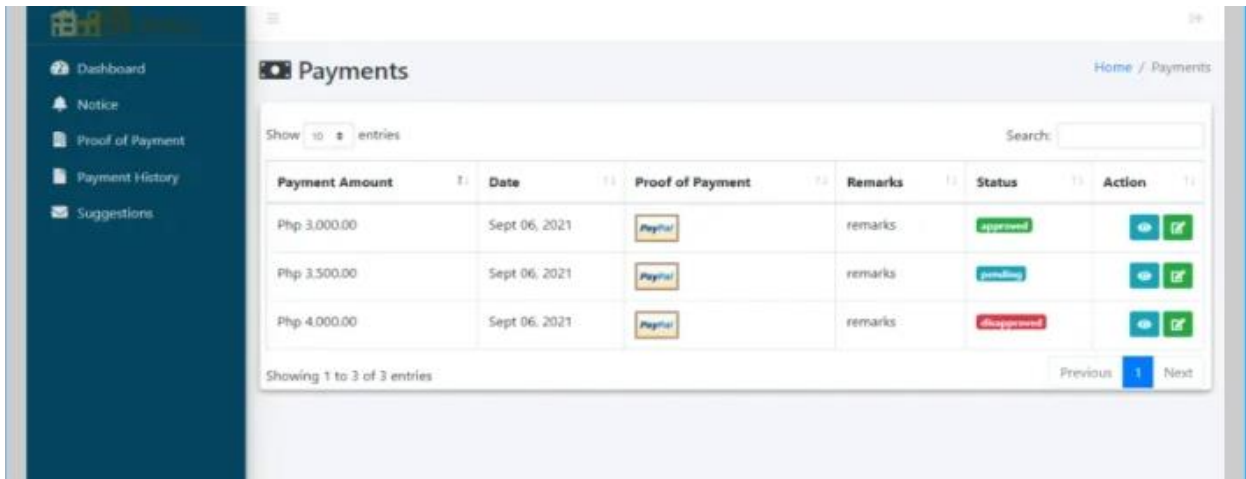


Fig. 9. Proof of payment form.

Figure 10. shows the data's reports of web-based boarding house and dormitory management systems. The figure shows that the system can store the boarder's information, such as rooms, and hold basic information on the different types of beds and their price. There will also view and monitor their transactions and payments. It also stores the payments and history of the customers.

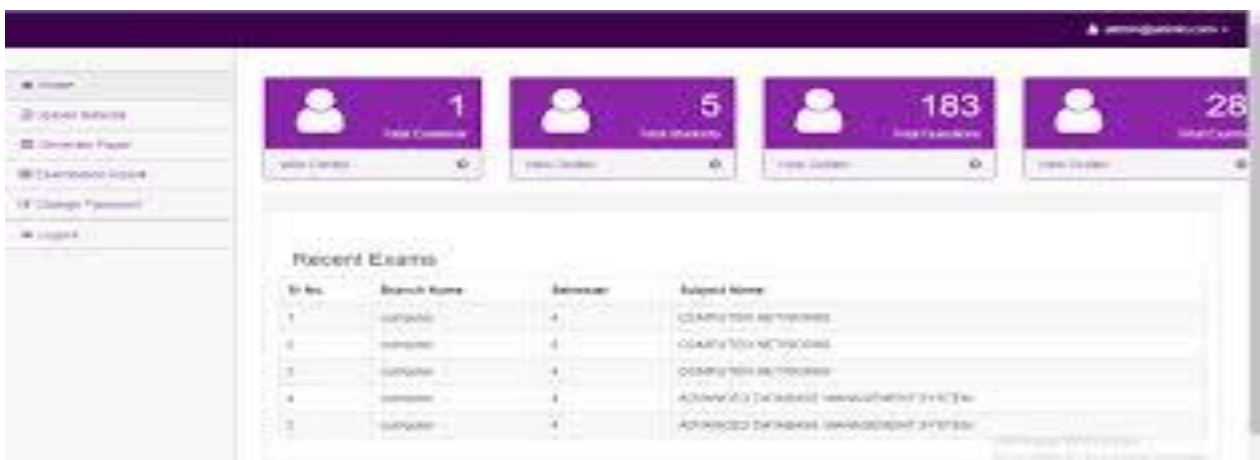


Fig. 10. Web Based Boarding-house and Dormitory Management System reports.

The image below shows the relationships of tables for Mamaco's Web-Based Boarding House and Dormitory Management System. As shown, the room's table will store information about the room, such as the room number, name, number of beds, and its room type or category. Then, the room type table will store information on the different types of rooms in the boarding house or Dormitory. The bed's table will hold information on the different types of beds and their price. Also, the customer's table will hold and store basic information of customers such as name, contact, address, and username and password. Also, the customers can log in with their accounts, and view and monitor their transactions and payments. On the other hand, the guardian's table will store information on the guardians of students that includes the name, contact, and address. The booking's table will record the booking transactions of customers, the table will store the customer info and bed information, and the started date and end date. Then, the record will be reviewed by the encoder or administrator to confirm the transaction. Moreover, the payment's table will store the payment transactions and history of customers. Users table will store the different users of the system.

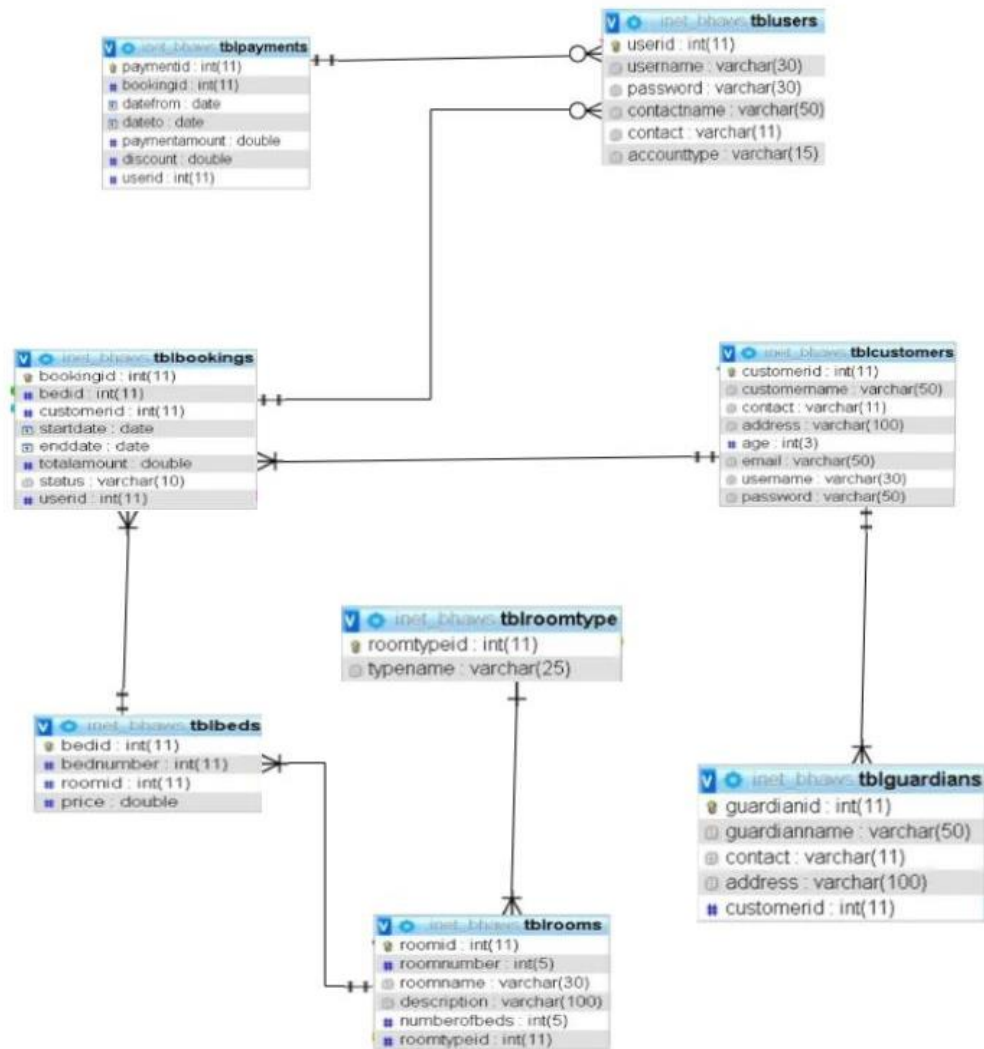


Fig. 16. Entity Relationship Diagram of Mamaco's Web-Based Boarding House and Dormitory Management System

se Case Diagram

Figure 11. shows the Use Case Diagram of Mamaco's Web-Based Boarding House and Dormitory Management System. It illustrated the activities on how the admin the customer interacts with the system. The administrator or employee in charge can only give customer accounts and update the information and reports. The administrator or employee in charge can also add, delete, and edit boarding house and dorm tenants' entries.

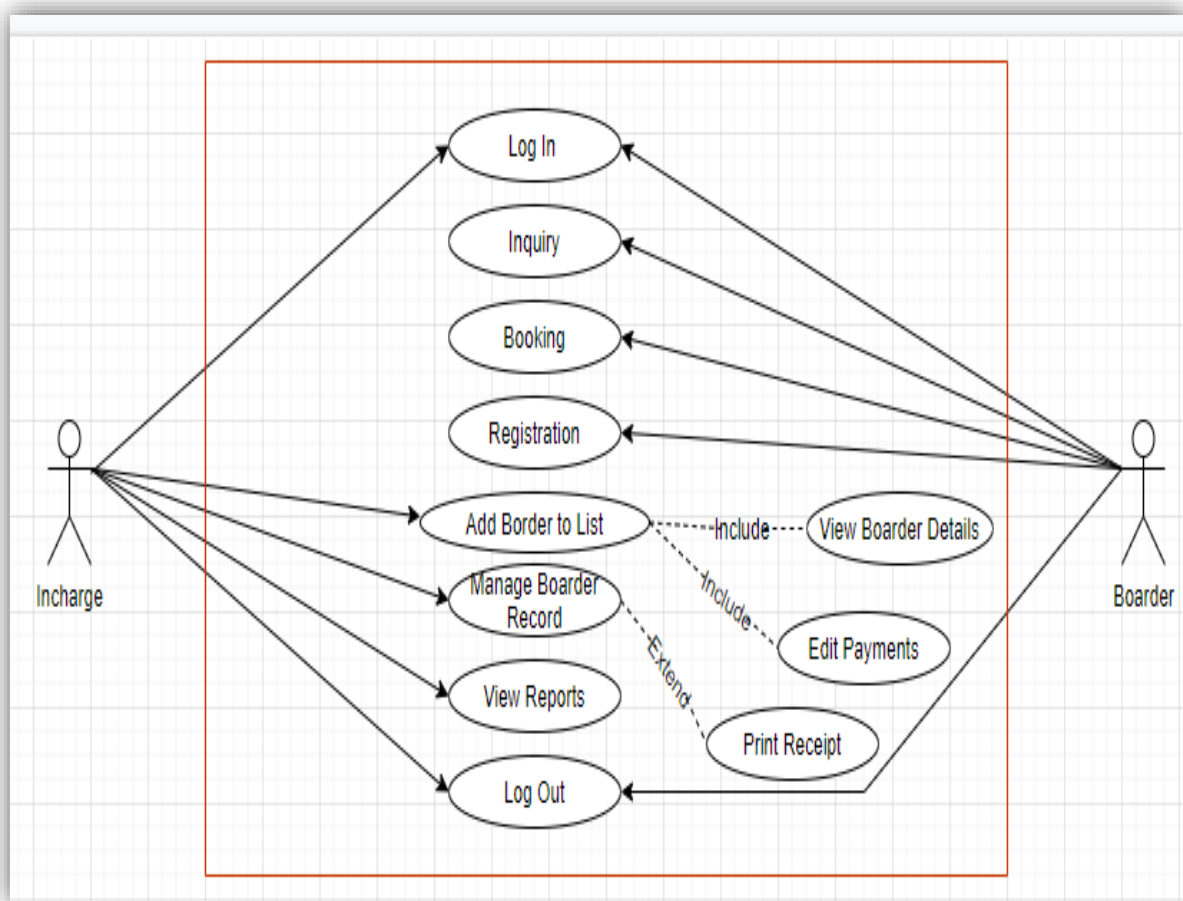


Fig. 17. Shows the Use Case Diagram of Mamaco's Web-Based Boarding House and Dormitory Management System.

Class Diagram

Figure 12. shows the class diagram for Mamaco's Web-Based Boarding House and Dormitory Management System. It is composed of six (6) fields, namely: the admin who oversees assisting the customer and managing all structures, customer that stands as boarder or client who is booking and paying the area, area for booking and paying, and guardians of boarders. Each class is indicated by one to many (1,1*).

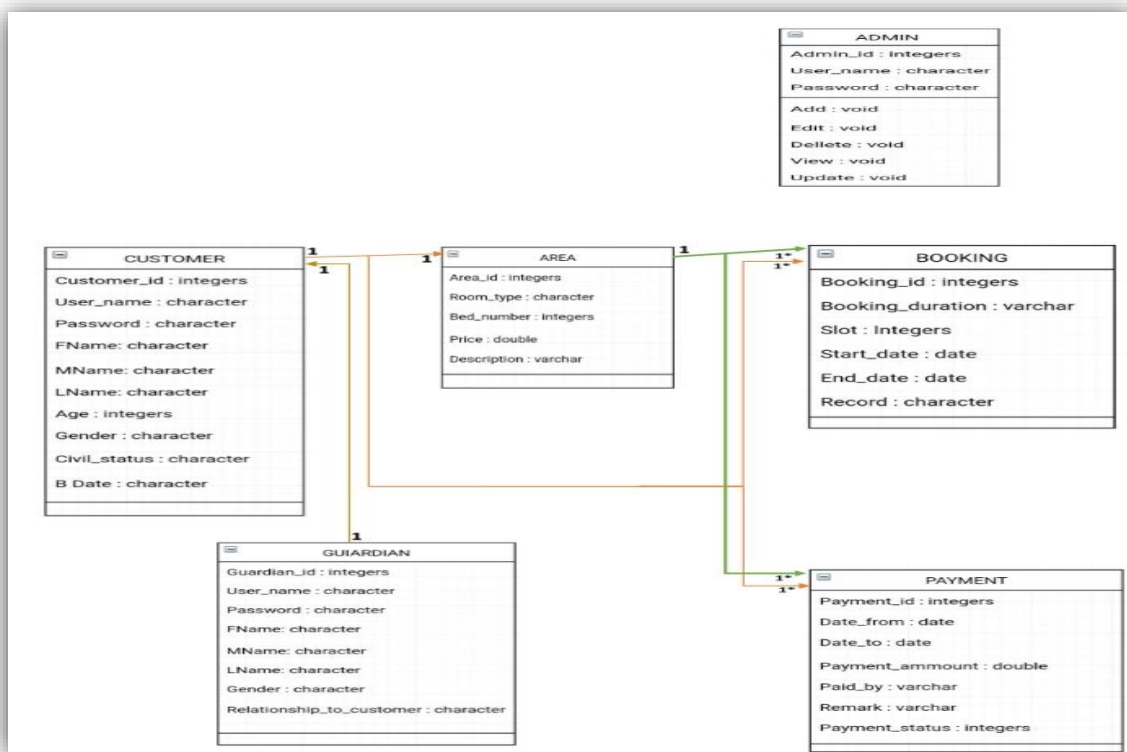


Fig.18. Class Diagram of Mamaco's Web-Based Boarding House and Dormitory Management System.

Activity Diagram

The following figures show users' activity, including the admin and the customer where they are required to login. All users can view rooms, beds, records, and payments, but only the admin or employee in charge can view the report module, print receipt, and edit payments. The admin or employee in charge can also add, edit, and delete members or boarders in the system. The client can inquire about the available services, register to be a renter, and book room and bed reservations. Figure 13. shows the activity diagram for login in a system. It models the actions that users will perform in the system to enter the system.

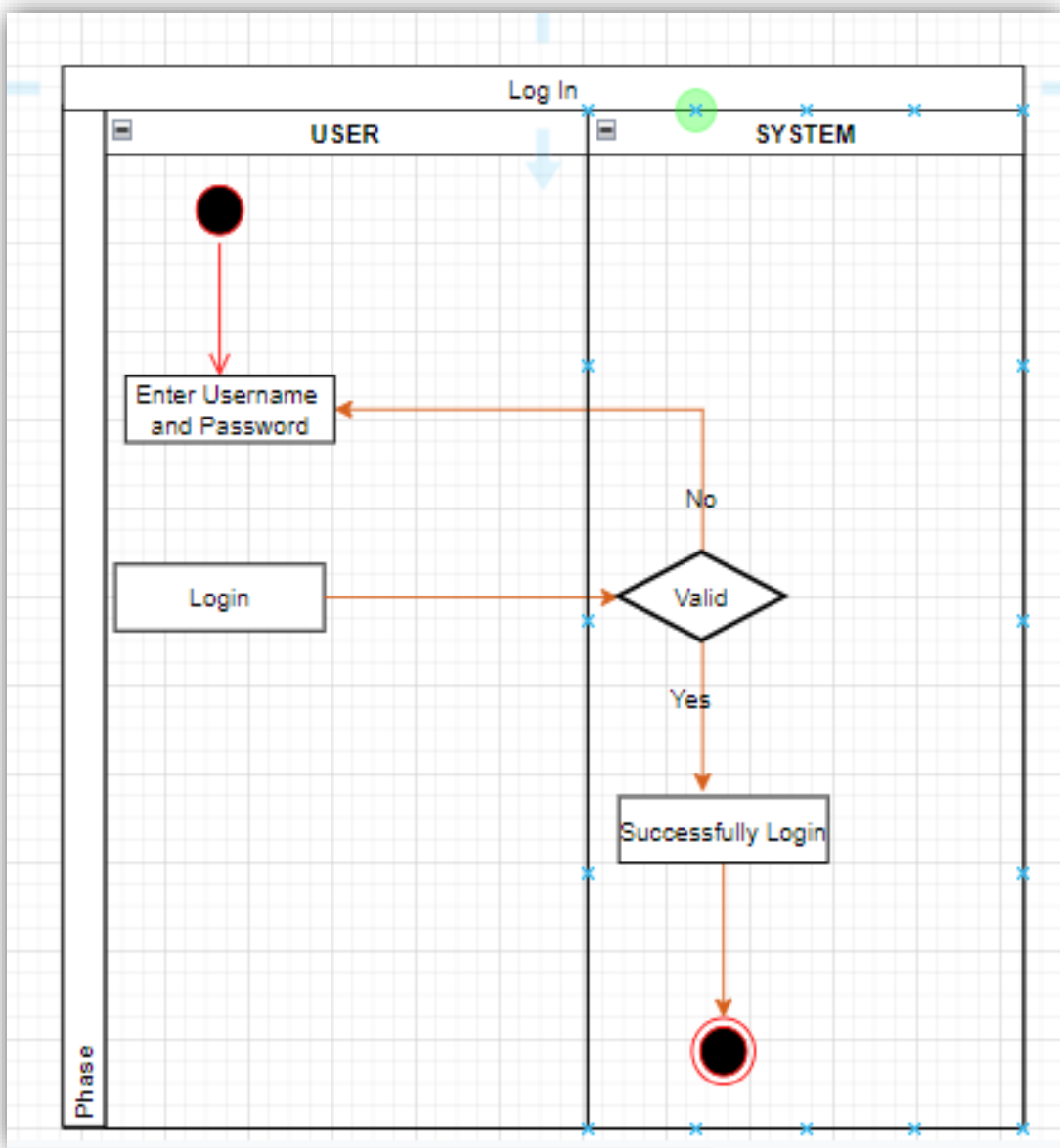


Fig. 19. The log in form of Mamaco's Web Based Boarding-house and Dormitory Management System.

Clients seek the establishment information, including the type of rooms, quality, price, and availability.

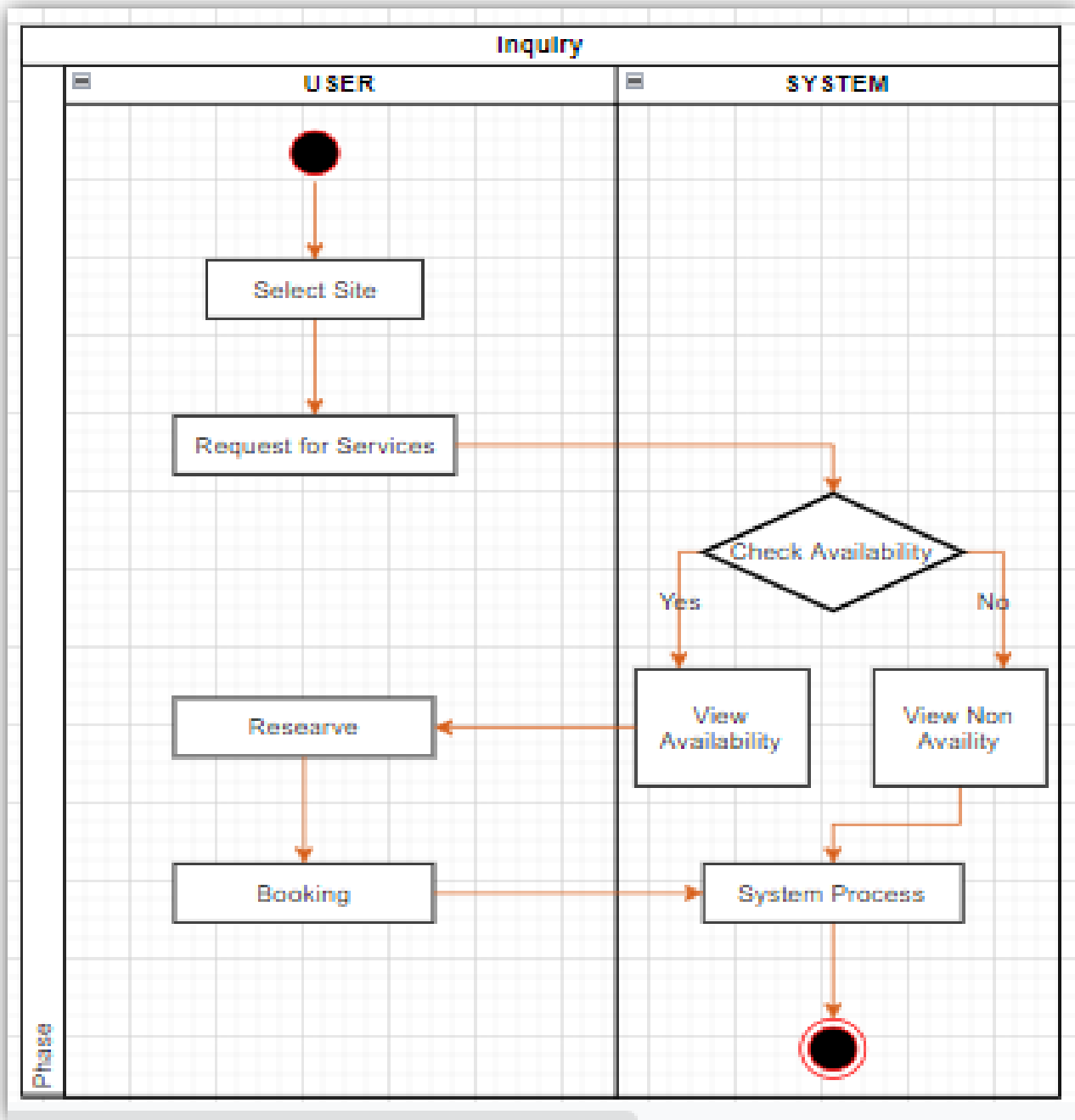


Fig. 20. Activity diagram for inquiry.

A process for boarder to reserve services.

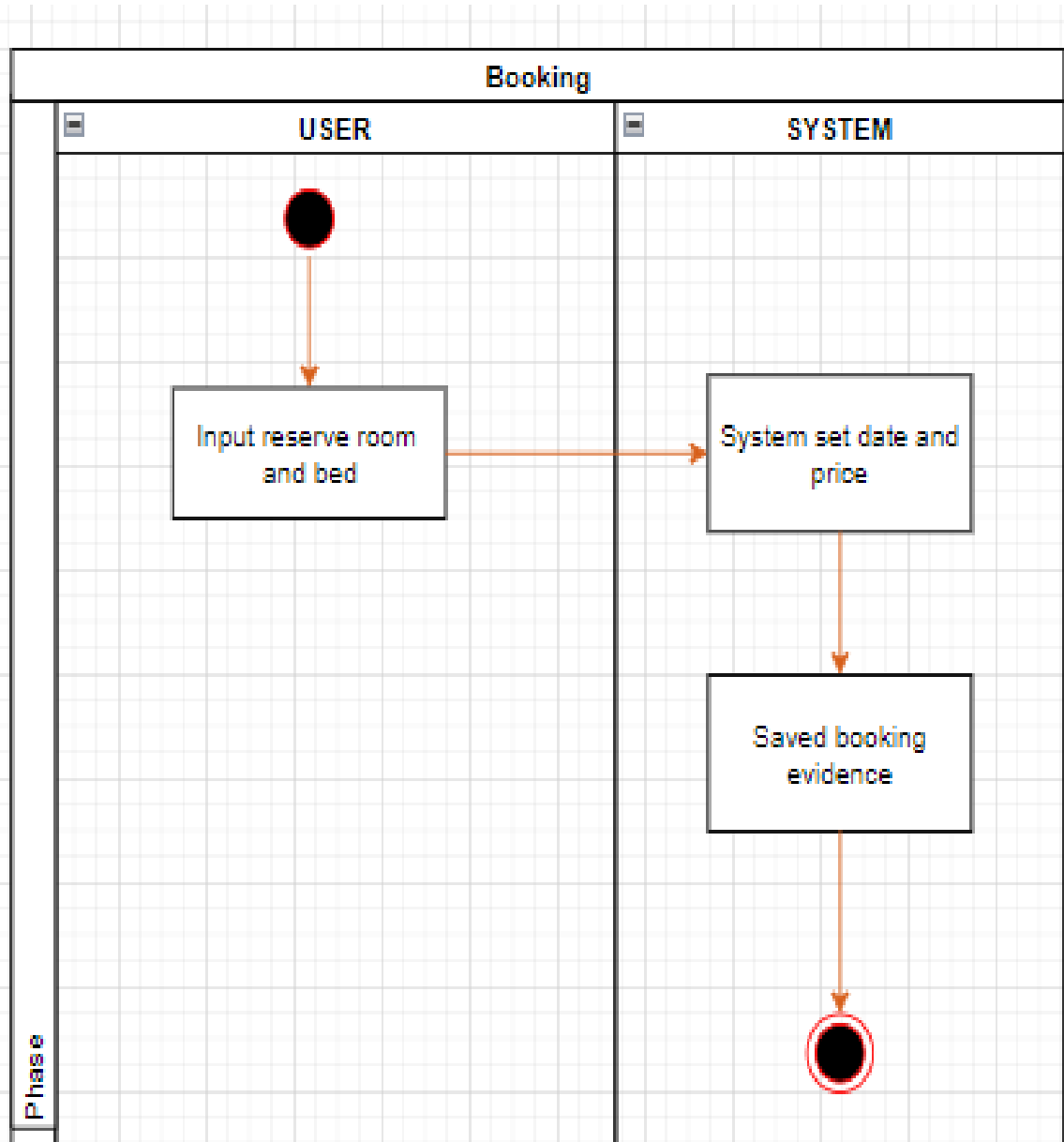


Fig. 21. Activity diagram for booking.

Registration is the process by which the establishment files required documents with the Securities. Then, the registration will able customers to see and choose room and bed availability.

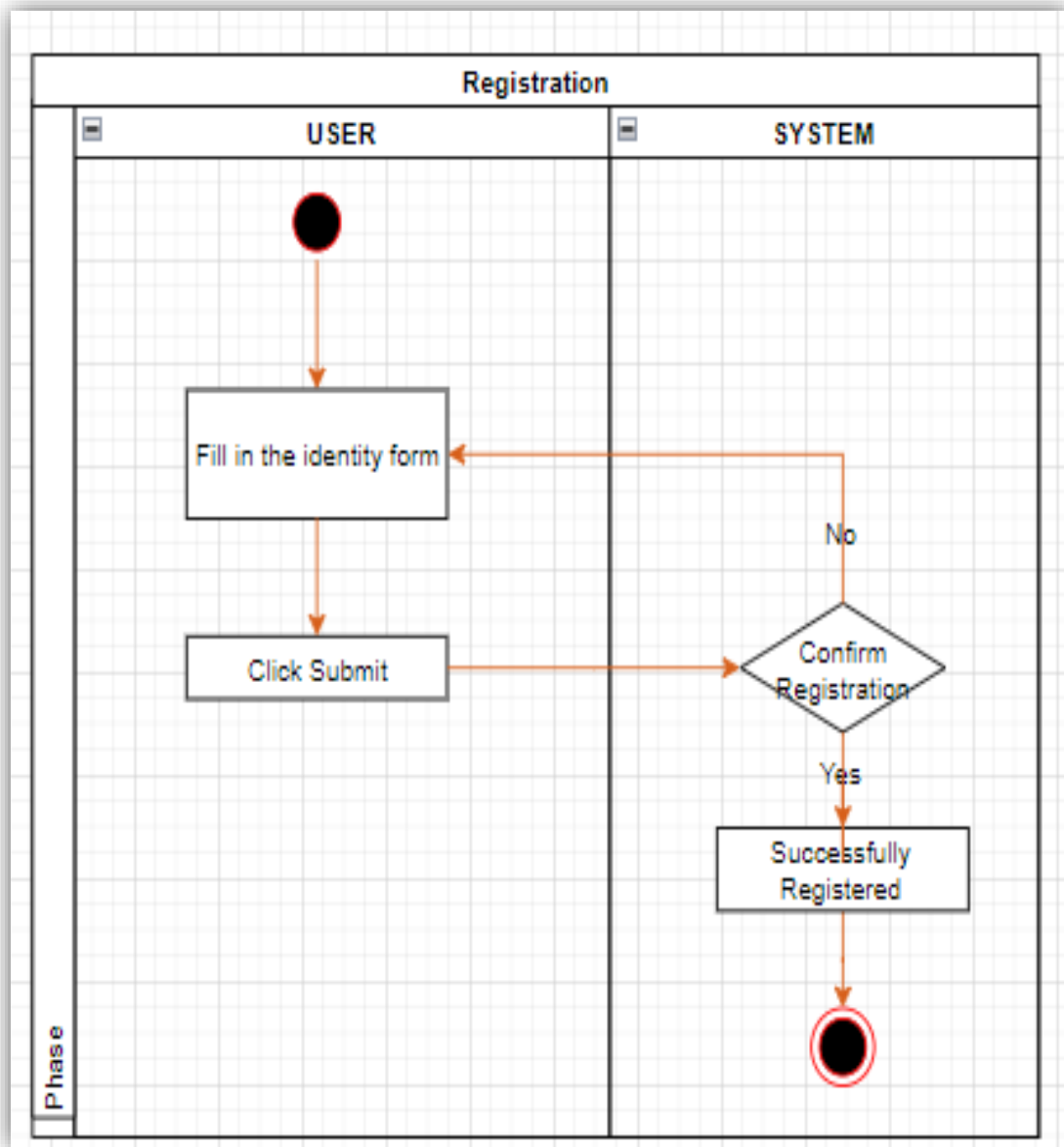


Fig. 22. Activity diagram for registration.

The process shows where the employee in charge adds the customer as a boarder or member of the boarding house. Then, the boarder record like personal information, room/bed, and payments will be viewed by the employee in charge.

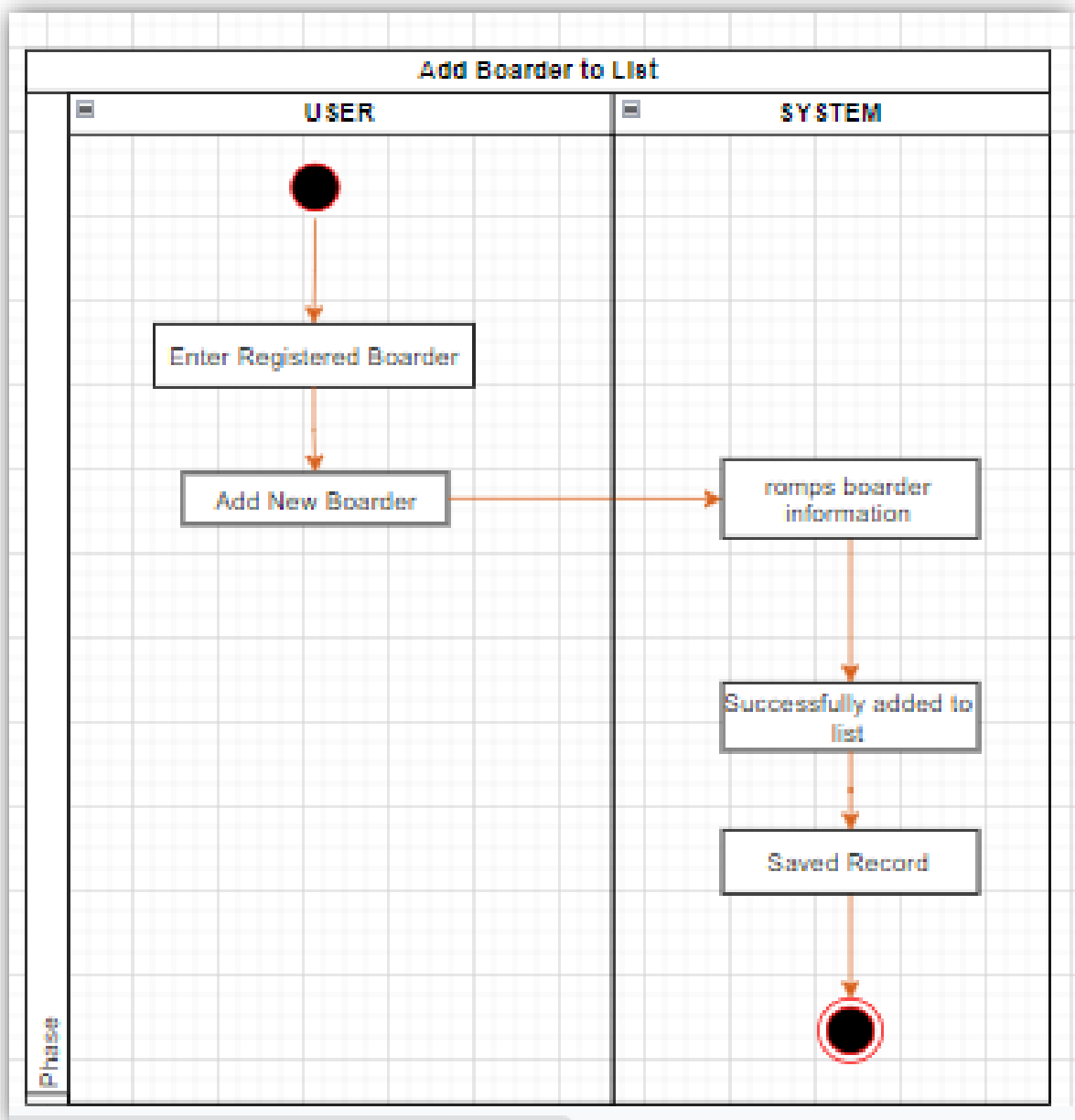


Fig. 23. Activity diagram for adding a boarder to list of tenants.

A process by viewing all transacted information between customer and employee in charge in the establishment.

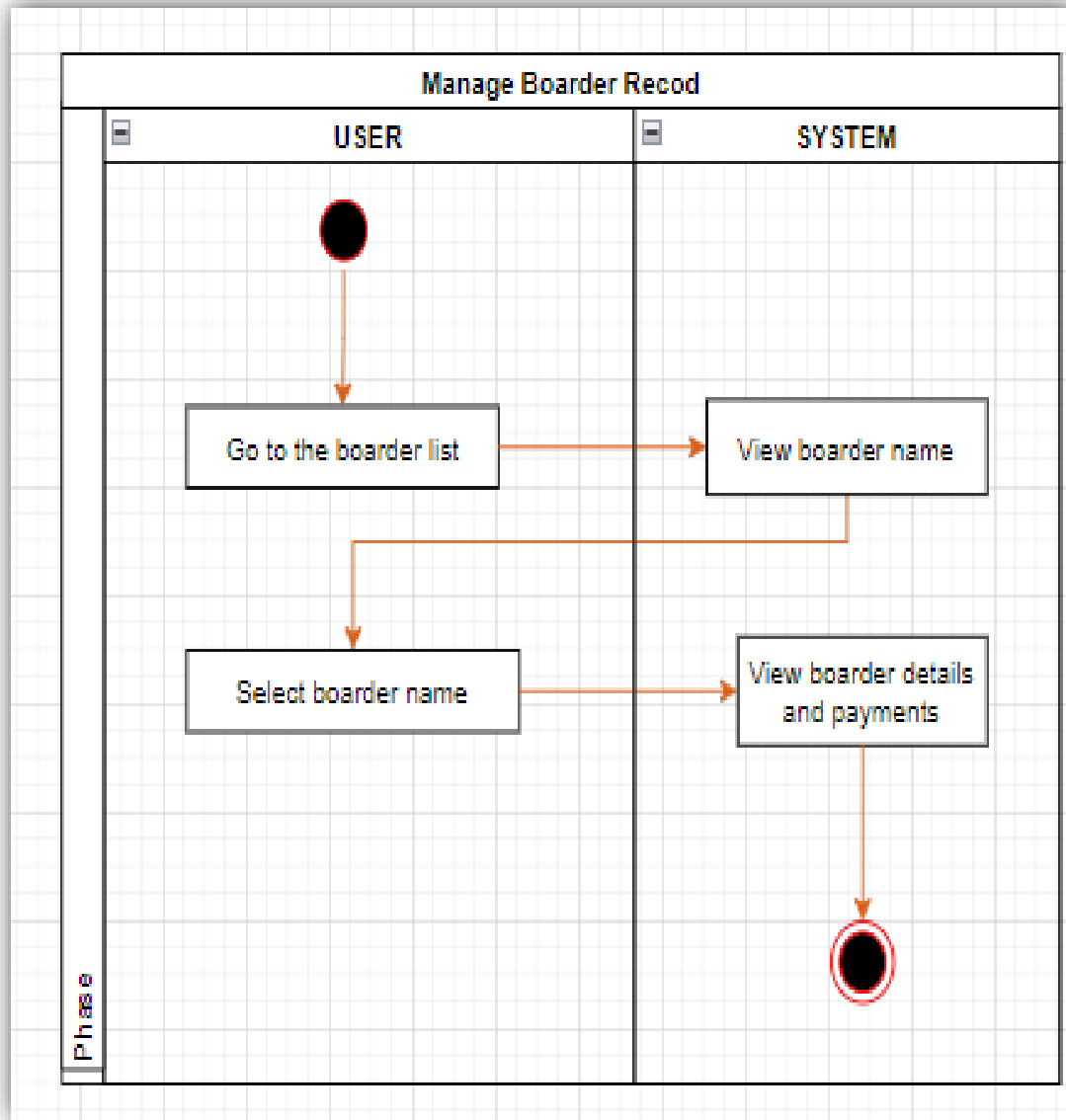


Fig. 24. Activity diagram for managing boarder record.

A process of viewing employees in charge about boarders' details. Also, the employees in charge can view the personal information of the boarder, payments, and guardians' information.

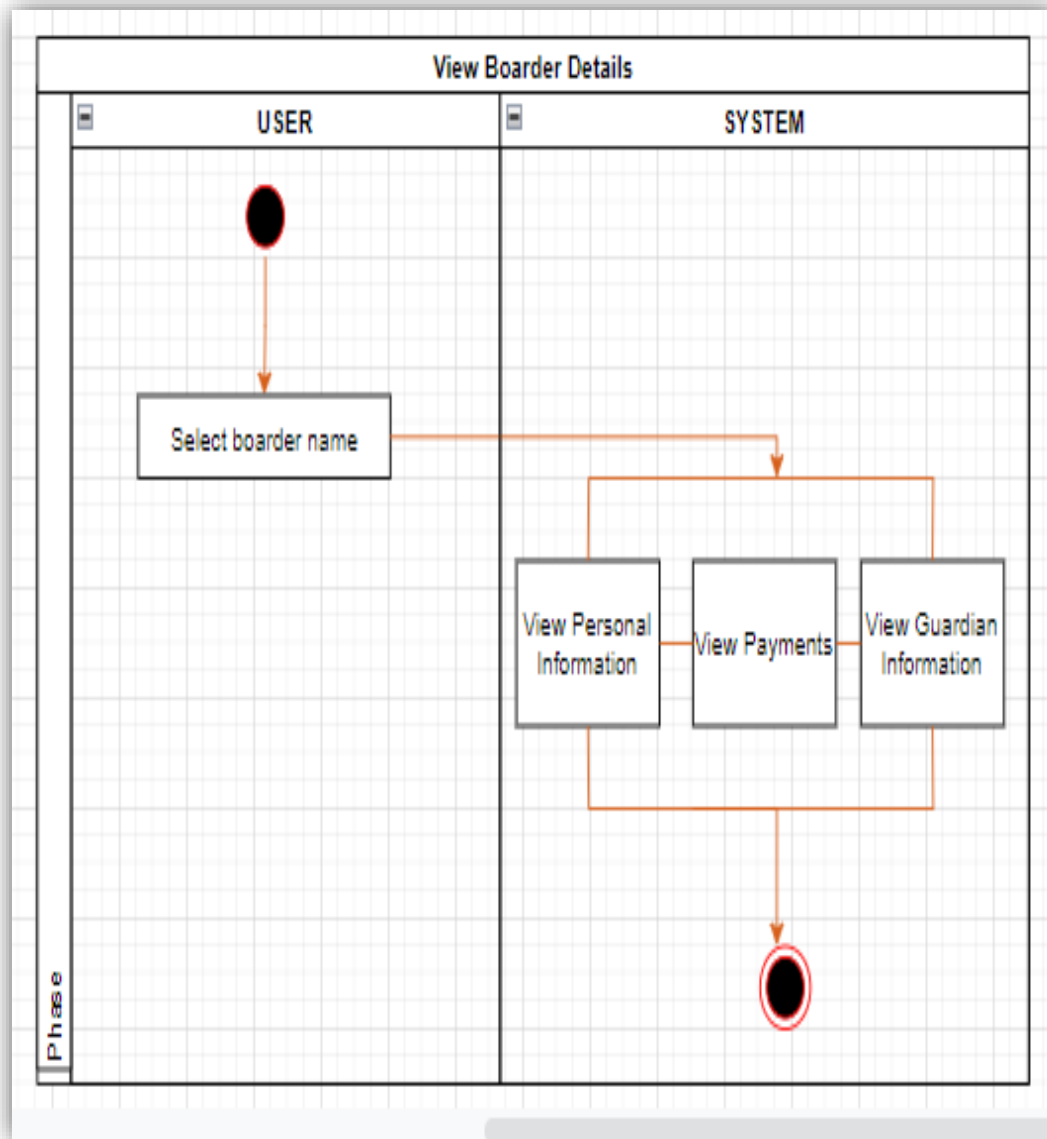


Fig 25. Activity diagram for viewing boarder details.

The image shows the processes of printing the employees' receipt in charge where all payment information is reflected.

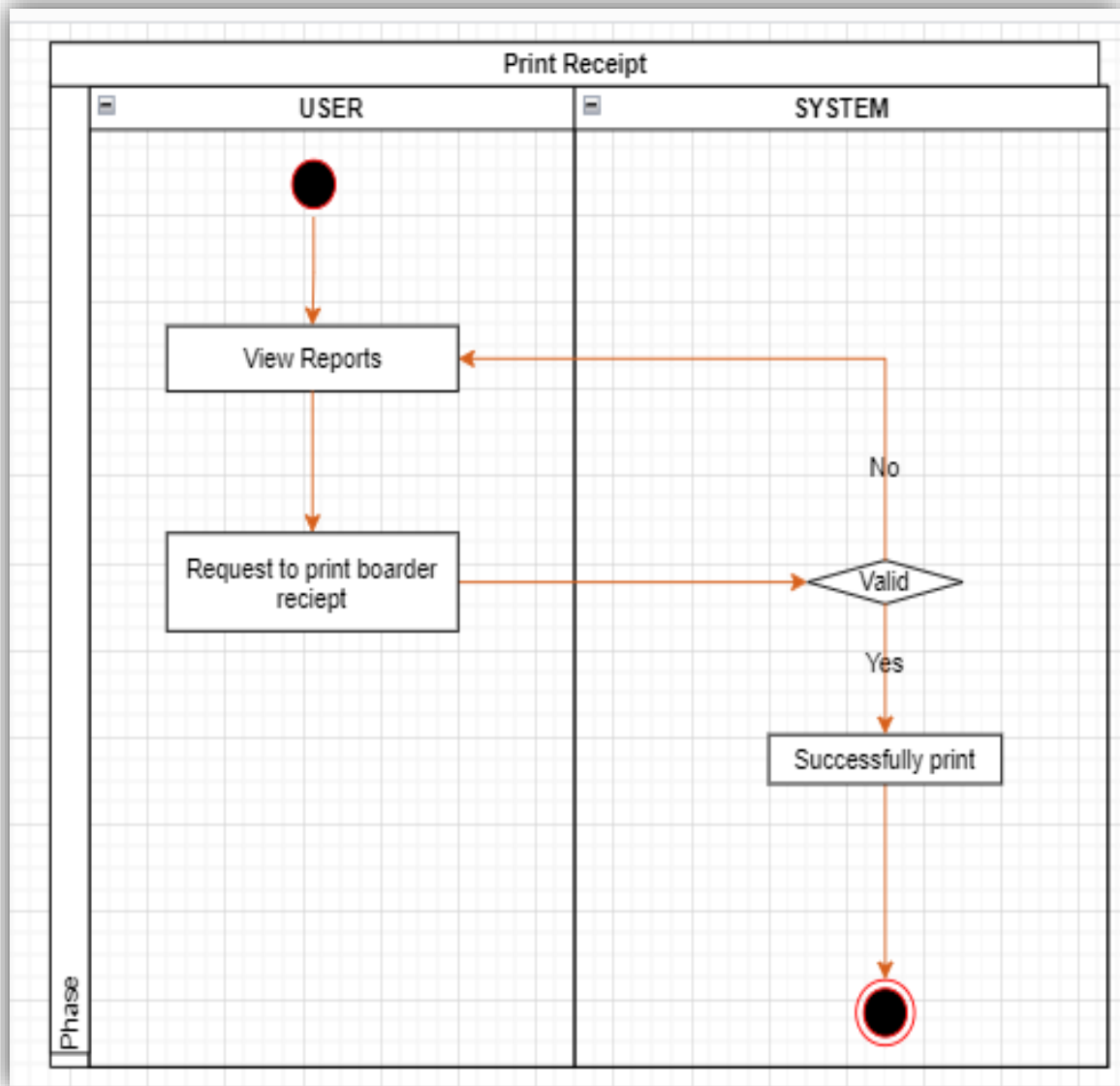


Fig. 26. Activity diagram for printing receipt.

It is a process of user to end access the website.

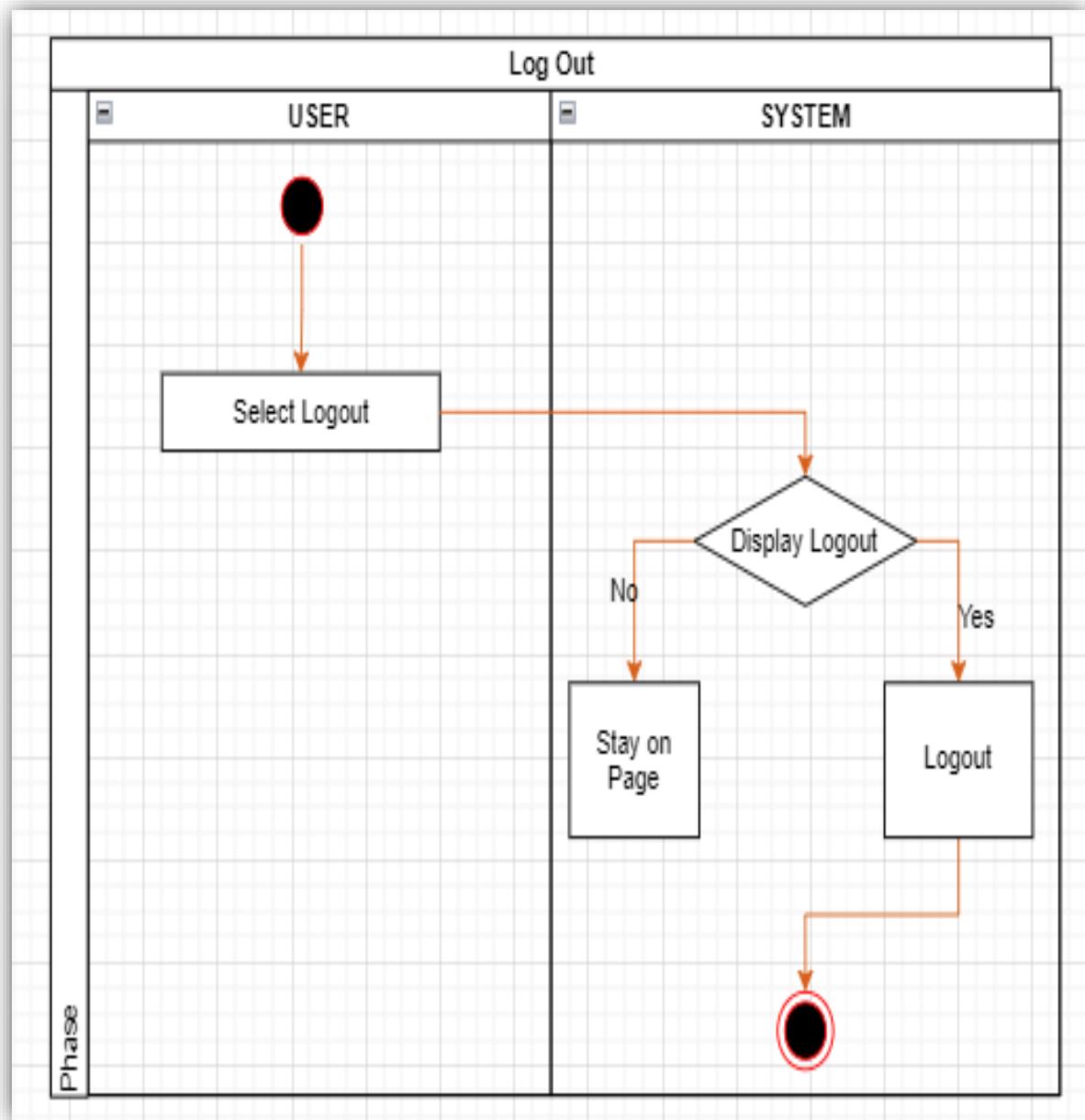


Fig. 27. Activity diagram for logout.

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APPENDICES

 UNIVERSITY OF SOUTHERN MINDANAO Kabacan, Cotabato Philippines 
APPLICATION FOR RESEARCH ADVISER

Date: 09-14-21

NOR AINE M. CORPUZ (Name of Proposed Adviser)
 Department of Library and Information Science
 College of CEIT(College of Engineering and Information Technology)

USM, Kabacan, Cotabato

Sir / Madam:

I would like to request that you will be my Research adviser effective 2nd semester,
 SY 2021 . I intend to work on Mamaco's Web Based Boarding House and Dormitory
Management System.

I am hoping for your most favorable approval on this request. Thank you very much.

Very truly yours,

ZUCRIA MAMACO NONAKAN
 Printed Name and Signature of Student

RECOMMENDING APPROVAL	NOTED
 <u>CATHERINE C. DAFFON</u> Department Research Coordinator	 <u>ELIZABETH R. GENOTIVA</u> Department Chairperson
<u>11-25-21</u> Date	<u>09-14-21</u> Date
APPROVED  <u>NOR AINE M. CORPUZ</u> Adviser <u>09-14-21</u> Date	

Appendix B. Application for Research Title

 UNIVERSITY OF SOUTHERN MINDANAO Kabacan, Cotabato Philippines 
APPLICATION FOR RESEARCH TITLE

Date: 09-14-21

ELIZABETH R. GENOTIVA
 Chairperson, Department of College of Engineering and Information System

SIR/MADAM:

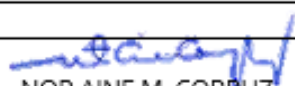
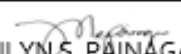
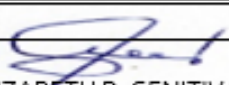
I would like to request your office to allow me to research on the study entitled
"MAMACO'S WEB BASED BOARDING HOUSE AND DORMITORY MANAGEMENT SYSTEM."

The study has the following objectives:

1. design and develop a module that will: manage records of the boarders; allow users to view room information online; show boarders bills; notify boarders' due date; and generate payment receipt.
2. create a centralized record for the management and customer as well.
3. evaluate the system in terms of user acceptability, effectiveness, functionality and productivity.

Very truly yours,


ZUCRIA MAMACO NONAKAN
 Printed Name and Signature of Student

NOTED	
 <u>NOR AINE M. CORPUZ</u> Adviser	<u>09-14-21</u> Date
 <u>CATHERINE C. DAFFON</u> Department Research Coordinator	<u>11-25-21</u> Date
 <u>MARILYN S. PAINAGAN</u> College Research Coordinator	<u>12-01-21</u> Date
REMARKS:	
APPROVED	
 <u>ELIZABETH R. GENOTIVA</u> Department Chairperson	<u>09-14-21</u> Date

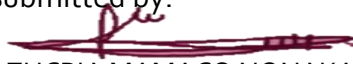
	UNIVERSITY OF SOUTHERN MINDANAO Kabacan, Cotabato Philippines	
ESTIMATED BUDGET OF THE RESEARCH		

Title of Study

MAMACO'S WEB BASED BOARDING HOUSE AND DORMITORY MANAGEMENT SYSTEM

ITEMS/DESCRIPTION	ESTIMATED COST
Adviser	00350.00
Earning Committee member	00300.00
Statistician	00100.00
Department Research Coordinator	00125.00
Travel	03,500.00
Internet Load	05,500.00
Computer Rent	05,000.00
Cost Prints	01,500.00
Sundry(Outline Hard bound)	000785.00
Grand Total	16,060.00

Prepared and submitted by:


 ZUCRIA MAMACO NONAKAN

Printed Name and Signature of the Student

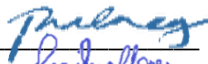
NOTED	
<u>NOR AINE M. CORPUZ</u> Adviser	12-04-21 Date
<u>CATHERINE C. DAFFON</u> Department Research Coordinator	Date
<u>ELIZABETH R. GENOTIVA</u> Department Chairperson	12-04-21 Date

Appendix C. Application for Thesis Outline Defense

	UNIVERSITY OF SOUTHERN MINDANAO Kabacan, Cotabato Philippines	
APPLICATION FOR THESIS OUTLINE DEFENSE		

Name	Zucria Mamaco Nonakan
Degree/Major	Bachelor of Science in Information System
Thesis Title	Mamaco's Web Based Boarding House and Dormitory Management System
Date of Examination	December 19, 2021
Time	4:00 pm
Place	Online

MEMBERS OF THE EXAMINING COMMITTEE

Name	Signature	Date
NELSON GEVEROLA BALNEG		<u>11-23-21</u>
DANILYN G. FLORES		<u>11-23-21</u>
CLARENCE DAVE GALAS		<u>11-23-21</u>

RECOMMENDING APPROVAL:


 NOR AINE M. CORPUZ

Adviser

Co-Adviser (Optional)

APPROVED:



CATHERINE C. DAFFON

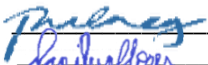
Department Research Coordinator

College Statistician
(Optional)

ELIZABETH R. GENITIVA

Department Chairperson

REPORT ON THE RESULT OF EXAMINATION

Name	Signature	Remarks
NELSON GEVEROLA BALNEG		<u>PASSED</u>
DANILYN G. FLORES		<u>PASSED</u>
CLARENCE DAVE GALAS		<u>PASSED</u>

APPROVED:



CATHERINE C. DAFFON

Department Research Coordinator

11-23-21

Date



April 17, 2021

Owner

Plang Village III, Kabacan, North Cotabato

Dear Sir/Madam:

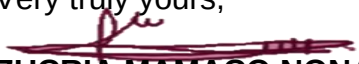
Greetings!

I, the graduating student of the University of Southern Mindanao, Kabacan, North Cotabato, who are taking up Bachelor of Science in Information System, would like to ask permission from your establishment to conduct survey that relating to my capstone project entitled “**Mamaco’s Web Based Boarding-house and Dormitory Management System**”. In addition, may I also request an interview about your establishment background, the difficulties and challenges you had encountered to manage and your clients information. By means of this survey, I will gain knowledge regarding factors helping the project in your establishment.

This is partial fulfilment of the requirements for the degree. Bachelor of Science in Information System.

I understand that I am carrying a huge responsibility. Rest assured that all information will be treated with utmost confidentiality. I hope that this request will merit your favorable approval. Thank you so much and Assalamualaykom!!!

Very truly yours,


ZUCRIA MAMACO NONAKAN
Researcher

CURRICULUM VITAE

ZUCRIA MAMACO NONAKAN

Poruk 2, Kitulaan, Carmen, North Cotabato

Cell no. 09355977619

Google Account: zucriamnonakan@usm.edu.ph

&zucriamnonakan@gmail.com



PERSONAL INTRODUCTION

Nick Name:	Beybie
Gender:	Female
Birthday:	March 09, 2001
Birth Place:	Doroluman, Arakan, North Cotabato
Nationality:	Filipino
Religion:	Islam
Father's Name:	Anok Ugalinan Nonakan
Mother's Name:	Noria Ballet mamaco
Siblings Number:	2 nd

EDUCATIONAL BACKGROUND

Tertiary:	University of Southern Mindanao Kabacan, Cotabato College of Engineering and Information Technology Bachelor of Science in Information System 2018 – 2022
Secondary Senior:	University of Southern Mindanao Kabacan, North Cotabato Senior High School Information and Communication Technology (ICT) 2017 – 2018

Secondary Junior:	Lumayong High School(LHS) Lumayong, Kayaga, Kabacan, North Cotabato 2012 – 2016
Elementary:	USM CED ILS Lumayong, Kayaga, Kabacan, North Cotabato Integrated Laboratory School 2006 –2012
Kinder Garden:	Lumayong Day Care Lumayong, Kayaga, Kabacan, North Cotabato 2004 – 2006

SEMINARS BACKROUND

University-Wide Software, Hardware And Document Freedom Day:	University of Southern Mindanao University Laboratory School(ULS) May 01, 2017
Group Growth Program:	University of Southern Mindanao Gymnasium 2018 – 2019
Free Open Source Software:	University of Southern Mindanao Gymnasium 2018 – 2019
Image Processing Using Python:	University of Southern Mindanao Gymnasium 2018 – 2019
USMMUSA Public Dawah:	University of Southern Mindanao Museum Park February 19, 2020
Public Da'wah:	University of Southern Mindanao

Museum Park
February 26, 2020

TRAININGS BACKGROUND

Training on University of Southern Mindanao
Community Service Gymnasium
And Earthquake May 24, 2019
And Fire Drill:

Training on University of Southern Mindanao
Standard Basic Gymnasium
First Aid Basic May 24, 2019
Water Safety:

Leadership University of Southern Mindanao
Training on Gymnasium
Climate Change May 24, 2019
And
Environmental
Preparedness:

2nd Young Leaders University of Southern Mindanao
Summit on Gymnasium
Sustainable May 24, 2019
Development
Goals(SDGs):

Values Bangsamoro Development Agency (BDA)
Transformation Aplal Function Hall
Training: May 20-22, 2021